Name: _____ Group members: _____

Investigation Planning Sheet

1. PLANNING

1.1 Aim

What are you trying to find out? What problem are you trying to solve?

_____

_____

_____

1.2 Variables

You can only change one variable at a time and measure its effect. Select the variable that you would like to investigate. The rest of the variables must be kept the same.

Change (Independent) **Measure (Dependent)** **Keep the Same (Controlled)**

_____ _____ _____

_____

_____

_____

1.3 Hypothesis

A statement about what you think will happen. This statement needs to link the independent and dependent variables.

_____

_____

_____

_____

2. CONDUCTING AND RECORDING

2.1 Equipment

Write a list of all the equipment you will need to conduct your experiment.

- _____
- _____
- _____
- _____
- _____
- _____
- _____
- _____

- _____
- _____

2.2 Procedure

Design a method by which you can slowly change one variable and measure its affect on the other.

1. _____
   _____
2. _____
   _____
3. _____
   _____
4. _____
   _____
5. _____
   _____
6. _____
   _____
7. _____
   _____
8. _____
   _____
9. _____
   _____
10. _____
    _____
    _____
11. _____
    _____
    _____

12. _____

_____

_____

2.3 Results
Construct a table so that you can record the changes in the variables.

	Temperature of air in greenhouse (^{0}C)				
Time (mins)	Air	Shape 1:	Shape 2:	Shape3:	Shape 4:
0					
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					

3. ANALYSING
3.1 Graphing
Draw a graph of your results. Use the following rules to Choose the type of graph.

Independent Data	**Dependent Data**	**Type of Graph**
Data is continuous and numerical (for example, time)	Data is continuous and numerical (for example, height)	Line graph
Data is in categories (for example, eye colour)	Data is numerical (for example, number of people)	Column graph

3.2 Analysing Your Results
Do your results support your hypothesis or not?

_____

_____

Explain **how** you used your results to come to your conclusion above.

_____

_____

_____

_____

4. EVALUATION

4.1 Errors

What are the possible sources of error in your investigation?

_____

_____

_____

How might these sources of error affect your data?

_____

_____

_____

4.2 Improving Your Investigation

What changes would you make to your investigation to improve your results?

_____

_____

_____

_____

_____

5. REPORT

You need to write a report of your experiment. Your report should include the following:

Aim: What is the aim or purpose of your investigation?

Hypothesis: Your prediction in the form of a statement. This needs to link your independent and dependent variables.

Equipment: A list of all the equipment you used in your investigation.

Diagram of Equipment: A labelled diagram of how you set up your equipment to conduct your investigation.

Method: A list of the steps you went through to conduct your investigation.

Results: The results of your investigation presented in both a table and a graph.

Discussion: Comment on whether or not your results supported your hypothesis, what were the possible sources of error in the investigation and what you could do to improve your investigation.

Conclusion: Short statement to sum up your experiment.